

Module 05: Action in Robotics

Learning Objectives

1. Define **Action** within the context of Robotics.
2. Analyze how Action interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Action.

Introduction

This module explores **Action**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Action is a critical component of the 8-part Active Inference spine, bridging the gap between Cognition and Learning.

Key Concepts

1. Action as a Markov Blanket Boundary

How does Action define the boundary between the agent and the environment?

2. Generative Models of Action

What parameters involved in Action must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Action drive the perception-action loop?

Applications

In Robotics, we see Action manifest in: * **Specific Example 1:** A cheetah-inspired quadruped robot generates galloping gaits through central pattern generators (CPGs) that act as the action

component of an Active Inference loop -- the CPG produces rhythmic motor predictions (expected joint trajectories), and ground reaction force sensors provide prediction error signals that modulate stride timing and foot placement in real time, allowing the robot to maintain stable high-speed locomotion over uneven terrain the same way biological quadrupeds use spinal reflexes to adapt gait without conscious planning. * **Specific Example 2:** An elephant-trunk-inspired continuum manipulator with 12 pneumatic muscle segments generates grasping actions by inflating and deflating antagonistic actuator pairs to minimize the difference between its predicted tip position (derived from a piecewise-constant-curvature generative model) and the observed tip pose from an electromagnetic tracker, embodying bio-inspired action where the compliant morphology itself absorbs uncertainty about object shape and contact geometry that would require explicit planning in a rigid-link manipulator.

Conclusion

Understanding Action allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.